Introduction to C++

Learning Objectives

After the completion of this chapter, the student will be able to

- Understand the basic building blocks of C++ programming language
- Able to construct simple C++ programs
- Execute and debug C++programs

9.1 Introduction

C++ is one of the most popular programming language which supports both procedural and Object Oriented Programming paradigms. Thus, C++ is called as a **hybrid language.** C++ is a superset (extension) of its predecessor C language. Bjarne Stroustrup named his new language as "C with Classes". The name C++ was coined by Rick Mascitti where ++ is the C language increment operator.

Bjarne Stroustrup
Inventor of C++ Programming Language



Bjarne is a Danish Computer Scientist born on 30th December 1950. He has a Master degree in Mathematics and Computer Science in 1975 from Aarhus University, Denmark and Ph.D in Computer Science in 1979 from the University of Cambridge, England.

History of C++

C++ was developed by Bjarne Stroustrup at AT & T Bell Laboratory during 1979. C++ is originally derived from C language and influenced by many languages like Simula, BCPL, Ada, ML, CLU and ALGOL 68. Till 1983, it was referred "New C" and "C with Classes". In 1983, the name was changed as C++ by Rick Mascitti.

Benefits of learning C++

- C++ is a highly portable language and is often the language of choice for multi-device, multi-platform app development.
- C++ is an object-oriented programming language and includes **classes**, **inheritance**, **polymorphism**, **data abstraction** and **encapsulation**.
- C++ has a rich function library.
- C++ allows exception handling, inheritance and function overloading which are not possible in C.
- C++ is a powerful, efficient and fast language. It finds a wide range of applications – from GUI applications to 3D graphics for games to real-time mathematical simulations.

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